# Hengming Jing

Biomedical optics and computational imaging

Shenzhen, China
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## Education

**Southern University of Science and Technology** Sep 2024 – Jun 2027 (expected)
Master's program in Electronic Science and Technology
Department of Biomedical Engineering | Advised by Prof. Jianbo Tang
GPA 3.44/4.00

**Hefei University of Technology** Sep 2020 – Jun 2024
B.Eng. in Mechanical Engineering
GPA 3.62/4.30 | Rank 34/349 (first six semesters)

## Research Experience

**M-Mode OCTA Readout** Mar 2026 – Present
*Manuscript in preparation*

- Developed an updated OCT angiography (OCTA) readout for M-mode optical coherence tomography (OCT) acquisitions.
- Evaluated small-vessel visibility and continuity against conventional M-OCTA at different repeat counts and in stitched volumes.

**3D OCTA Vessel Segmentation and Analysis** Feb 2025 – Present
*Manuscript in preparation*

- Fine-tuned VesselFM on public vessel data, then adapted it to local OCTA with manually screened and corrected pseudo-labels.
- Extracted vessel centerlines and built graphs from 13 local OCTA volumes to measure segment length, tortuosity, and apparent caliber.

**Deep-UV Imaging and Automated Acquisition** Apr 2025 – Mar 2026
*Manuscript in preparation*

- Co-built and aligned a 255 nm deep-UV imaging module on a serial-sectioning platform and integrated its illumination, camera, and motion stage.
- Developed Python/PyQt controls to synchronize image acquisition with stage motion.
- Automated focus-stack processing, illumination correction, and stitching for 3D tissue visualization.

**Photoacoustic Imaging**
Helped assemble and tune a photoacoustic imaging system, then ran imaging experiments.

## Presentations & Honors

**Best Student Paper Award, PIBM 2026** Jul 2026
Oral presentation on OCT-based 3D multiparametric imaging of the cerebral microvasculature at the International Conference on Photonics and Imaging in Biology and Medicine.

**iCAN Innovation Contest, Anhui Division Third Prize** 2022

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Curriculum Vitae

## Publications

### Journal articles and invited review

J. Tang, **H. Jing**, J. Yang, and W. Chen.
Dynamic optical coherence tomography: from qualitative to quantitative (invited review).
*Chinese Journal of Lasers* **53**(9), 0907106 (2026). DOI

X. Zhai, **H. Jing**, H. Chen, S. Chang, and J. Tang.
3D reconstruction of murine hepatic vasculature via serial sectioning SS-OCT imaging.
*Journal of Innovative Optical Health Sciences* **19**(6), 2650006 (2026). DOI

F. Yang, C. Chen, L. Meng, **H. Jing**, X. Liu, W. Chen, S.-C. Chen, and J. Tang.
Learned feature-based clustering of brain tissue with multimodal optical coherence tomography and photoacoustic elastography.
*Optics Letters* **51**(3), 796–799 (2026). DOI

F. Yang, **H. Jing**, S. Chang, H. Sun, W. Chen, and J. Tang.
Multimodal label-free imaging of unprocessed biological tissues using ultraviolet photoacoustic elastography and optical coherence microscopy.
*Optics Letters* **50**(16), 5010–5013 (2025). DOI

### Conference paper

J. Zhou, **H. Jing**, M. Li, and J. Tang.
OCT-based three-dimensional multi-parametric imaging of the cerebral microcirculation.
*European Conferences on Biomedical Optics (ECBO)*, M1C.2 (2025). DOI

## Industry Research Experience

**Software Development Intern, Raymind** 2026
Image analysis | Shenzhen, China

- Designed a workflow that combined bright-field and fluorescence features, clustered cells within each acquisition, and assigned leukocyte classes using reviewed examples.
- Reduced mean absolute error in three leukocyte proportions from 16.07 to 4.01 percentage points across 111 acquisition groups compared with the company baseline. Same-day complete blood counts served as the reference.

## Technical Skills

**Computational imaging** Image registration, 3D segmentation, vascular graph analysis.

**Experimental imaging** Instrument assembly, optical alignment, camera triggering, and imaging experiments.

**Software** Python, PyTorch, MATLAB, PyQt, VTK, ParaView, and SolidWorks.